SEQUENCES OF EVENTS

On the accompanying map you should be able to identify (a) two discordant, parallel-sided intrusions (dykes), (b) an unconformity, and (c) a fault. Make sure that you can. Then complete the following.

1. Draw strike-lines on the unconformity. Use these to establish:
   (a) the strike/dip and dip/dip direction of the unconformity;
   (b) whether or not the unconformity pre-dates or post-dates the fault.

2. Draw strike-lines on the limestone/mudstone contact and (in a different colour) the 200 ft and 100 ft strike-lines on the conglomerate/limestone contact. Use these to establish:
   (a) the strike/dip and dip/dip direction of the limestone/mudstone contact;
   (b) whether or not the conglomerate/limestone contact is parallel to the limestone/mudstone contact;
   (c) the stratigraphic sequence of sedimentary units;
   (d) the thickness of the limestone.

3. Examine the two dykes and answer the following:
   (a) Which dyke is younger?
   (b) What is the age of each intrusion relative to the sedimentary units?
   (c) By drawing strike-lines, determine the orientation (strike dip and dip/dip direction) of each intrusion.

4. Examine the fault and answer the following:
   (a) What is the age of the fault relative to the sedimentary and igneous units?
   (b) What is the orientation (strike/dip) of the fault?

5. Draw a cross section in the plane of the fault, looking north. Make sure that the vertical scale is the SAME as the horizontal scale. Fully annotate the cross section and draw a stratigraphic column to scale as a key to the units. OPTIONAL: what is the scale written in the format 1:10000?

6. OPTIONAL: Determine the fault slip vector and movement sense by constructing a juxtaposition section. You only need to show the limestone/mudstone contact (north and south of the fault) and the west side of the ore vein (north and south of the fault) to obtain the answers.

7. Write a sequence of geological events, briefly noting the evidence used to obtain this sequence, and drawing attention to any ambiguities in age relationships that there might be.